

Canalicular method comparison: current against experimental

=====

PIXEL SIZE 0.13 um/px. PIXEL SIZE 0.13 UM/PIX, ROUNDED, UNCONFIRMED.

PRE-VALIDATION: no number here has been checked against a manual count.

Both methods are measured by src/quantification.py, the same code.

No claim is made that either method is better.

Measures 1 to 6 of 34

image	method	lacuna_count	interior_lacuna_count	n_bridges	field_length_density_per_px	field_length_density_um_per_um2	total_skeleton_length_px
542_z06	current	16	13	17	0.03465	0.266539	34964
542_z06	ridge	16	13	18	0.0386429	0.297253	38993
542_z18	current	12	11	14	0.0340728	0.262098	34325
542_z18	ridge	12	11	12	0.0373694	0.287457	37646
543-2	current	12	12	21	0.0447014	0.343857	45785
543-2	ridge	12	12	10	0.0440023	0.33848	45069
543_3	current	10	9	14	0.0451398	0.347229	46420
543_3	ridge	10	9	13	0.0448899	0.345307	46163
543_z13	current	11	11	19	0.0438024	0.336942	45033
543_z13	ridge	11	11	20	0.0429066	0.330051	44112
682_z08	current	10	8	24	0.0346465	0.266512	35267
682_z08	ridge	10	8	22	0.0368265	0.283281	37486
682_z23	current	14	11	19	0.0382982	0.294601	39168
682_z23	ridge	14	11	14	0.0414603	0.318926	42402
682_z29	current	13	11	17	0.0374846	0.288343	38368
682_z29	ridge	13	11	21	0.039548	0.304215	40480

Measures 7 to 12 of 34

image	method	total_skeleton_length_um	roots_per_cell	ring_length_r30_px_per_cell	ring_length_r30_um_per_cell	ring_length_r60_px_per_cell	ring_length_r60_um_per_cell
542_z06	current	4545.32	8.6154	361.462	46.99	907.385	117.96
542_z06	ridge	5069.09	11	426.154	55.4	1029.69	133.86
542_z18	current	4462.25	8.2727	357.091	46.4218	968.727	125.935
542_z18	ridge	4893.98	8.5455	394.454	51.2791	1052.27	136.795
543-2	current	5952.05	7.5833	306.583	39.8558	966.417	125.634
543-2	ridge	5858.97	6.9167	289.333	37.6133	949.75	123.468
543_3	current	6034.6	6.8889	313.556	40.7622	989.889	128.686
543_3	ridge	6001.19	7.1111	296.667	38.5667	959.444	124.728
543_z13	current	5854.29	6.7273	302.909	39.3782	961.364	124.977
543_z13	ridge	5734.56	6.3636	303.091	39.4018	957.455	124.469
682_z08	current	4584.71	10	396.75	51.5775	982	127.66
682_z08	ridge	4873.18	10.625	427.75	55.6075	1039.62	135.151
682_z23	current	5091.84	6.7273	283.364	36.8373	795.182	103.374
682_z23	ridge	5512.26	6.8182	310.909	40.4182	864.455	112.379
682_z29	current	4987.84	5.5455	250.273	32.5355	697.455	90.6691
682_z29	ridge	5262.4	5.3636	265	34.45	727.909	94.6282

Measures 13 to 18 of 34

image	method	width_median_px	width_median_um	width_p10_px	width_p10_um	width_p90_px	width_p90_um
542_z06	current	6.3246	0.822198	4	0.52	8	1.04
542_z06	ridge	8	1.04	4	0.52	10	1.3
542_z18	current	6	0.78	4	0.52	8	1.04
542_z18	ridge	8	1.04	4	0.52	10	1.3
543-2	current	6	0.78	2.8284	0.367692	8	1.04
543-2	ridge	7.2111	0.937443	4	0.52	10	1.3
543_3	current	6	0.78	2.8284	0.367692	8	1.04
543_3	ridge	7.2111	0.937443	4	0.52	10	1.3

543_z13	current	6	0.78	2.8284	0.367692	8	1.04
543_z13	ridge	7.2111	0.937443	4	0.52	8.9443	1.16276
682_z08	current	6	0.78	2.8284	0.367692	8	1.04
682_z08	ridge	7.2111	0.937443	4	0.52	10	1.3
682_z23	current	6	0.78	4	0.52	8	1.04
682_z23	ridge	7.2111	0.937443	4	0.52	10	1.3
682_z29	current	6	0.78	2.8284	0.367692	8	1.04
682_z29	ridge	7.2111	0.937443	4	0.52	10	1.3

Measures 19 to 24 of 34							
image	method	width_pixels_used	ring_width_mean_r30_px_per_cell	ring_width_mean_r30_um_per_cell	junction_count	junction_density_per_px2	junction_density_per_um2
542_z06	current	33745	5.7445	0.746785	375	0.00037163	0.022
542_z06	ridge	37711	6.6889	0.869557	782	0.00077498	0.0459
542_z18	current	32892	5.5062	0.715806	395	0.0003921	0.0232
542_z18	ridge	36103	6.5745	0.854685	777	0.00077129	0.0456
543-2	current	45758	4.9642	0.645346	681	0.00066488	0.0393
543-2	ridge	45049	6.3687	0.827931	998	0.00097438	0.0577
543_3	current	46400	5.0947	0.662311	698	0.00067875	0.0402
543_3	ridge	46139	6.4428	0.837564	1031	0.00100257	0.0593
543_z13	current	45013	4.9871	0.648323	643	0.00062543	0.037
543_z13	ridge	44075	6.0276	0.783588	904	0.0008793	0.052
682_z08	current	32998	5.5385	0.720005	480	0.00047155	0.0279
682_z08	ridge	35245	6.2245	0.809185	784	0.00077021	0.0456
682_z23	current	36498	5.2823	0.686699	481	0.00047032	0.0278
682_z23	ridge	39569	6.1843	0.803959	913	0.00089272	0.0528
682_z29	current	34940	5.0813	0.660569	485	0.00047383	0.028
682_z29	ridge	37002	5.9558	0.774254	820	0.00080112	0.0474

Measures 25 to 30 of 34							
image	method	thread_end_fraction	connected_to_lacuna_fraction	mean_edge_length_px	mean_edge_length_um	median_edge_length_px	median_edge_length_um
542_z06	current	0.803952	0.374259	31.5991	4.10788	21.3137	2.77078
542_z06	ridge	0.59898	0.693801	25.2639	3.28431	17	2.21
542_z18	current	0.800303	0.299316	30.7951	4.00336	23.0355	2.99462
542_z18	ridge	0.599796	0.657465	24.2408	3.1513	16.8284	2.1877
543-2	current	0.723182	0.407944	27.4088	3.56314	19.364	2.51731
543-2	ridge	0.52304	0.8344	24.2457	3.15194	17.4853	2.27309
543_3	current	0.722462	0.257258	30.6592	3.9857	22.8284	2.9677
543_3	ridge	0.527473	0.795191	24.3754	3.1688	17.6569	2.29539
543_z13	current	0.728948	0.386413	26.8926	3.49604	18.6569	2.42539
543_z13	ridge	0.563401	0.685067	25.0762	3.25991	17.5711	2.28424
682_z08	current	0.777315	0.409921	25.4567	3.30937	17.4142	2.26385
682_z08	ridge	0.608652	0.695992	22.7178	2.95331	16.0711	2.08924
682_z23	current	0.781448	0.306746	32.7335	4.25535	21.5563	2.80233
682_z23	ridge	0.583333	0.699742	23.2194	3.01852	16.7279	2.17463
682_z29	current	0.775265	0.246415	38.2007	4.96609	21.1924	2.75501
682_z29	ridge	0.595191	0.690966	24.9932	3.24912	17.8284	2.3177

Measures 31 to 34 of 34					
image	method	owned_length_px_per_cell	owned_length_um_per_cell	analysed_area_px2	analysed_area_um2
542_z06	current	988.281	128.477	1.00906e+06	17053.1
542_z06	ridge	2129.09	276.782	1.00906e+06	17053.1
542_z18	current	972.615	126.44	1.0074e+06	17025.1
542_z18	ridge	2386	310.18	1.0074e+06	17025.1
543-2	current	1687.33	219.353	1.02424e+06	17309.7
543-2	ridge	3496.39	454.531	1.02424e+06	17309.7
543_3	current	1435.15	186.569	1.02836e+06	17379.3

543_3	ridge	4542.55	590.532	1.02836e+06	17379.3
543_z13	current	1713.99	222.818	1.02809e+06	17374.8
543_z13	ridge	3061.43	397.987	1.02809e+06	17374.8
682_z08	current	1860.56	241.873	1.01791e+06	17202.7
682_z08	ridge	3413.05	443.697	1.01791e+06	17202.7
682_z23	current	1065.82	138.556	1.02271e+06	17283.8
682_z23	ridge	2387.95	310.433	1.02271e+06	17283.8
682_z29	current	848.515	110.307	1.02357e+06	17298.3
682_z29	ridge	2592.49	337.024	1.02357e+06	17298.3

Plausibility, reporting only

image	method	measure	measured	low	high	unit	verdict
542_z06	current	canalicular diameter	0.822198	0.1	0.7	um	above
542_z06	current	length per canaliculus	14.9124	25	50	um	below
542_z06	current	canaliculi per lacuna	8.6154	40	115	count	below
542_z06	current	areal density	0.266539	0.5	0.9	per um^2	below
542_z06	ridge	canalicular diameter	1.04	0.1	0.7	um	above
542_z06	ridge	length per canaliculus	25.162	25	50	um	inside
542_z06	ridge	canaliculi per lacuna	11	40	115	count	below
542_z06	ridge	areal density	0.297253	0.5	0.9	per um^2	below
542_z18	current	canalicular diameter	0.78	0.1	0.7	um	above
542_z18	current	length per canaliculus	15.284	25	50	um	below
542_z18	current	canaliculi per lacuna	8.2727	40	115	count	below
542_z18	current	areal density	0.262098	0.5	0.9	per um^2	below
542_z18	ridge	canalicular diameter	1.04	0.1	0.7	um	above
542_z18	ridge	length per canaliculus	36.2974	25	50	um	inside
542_z18	ridge	canaliculi per lacuna	8.5455	40	115	count	below
542_z18	ridge	areal density	0.287457	0.5	0.9	per um^2	below
543-2	current	canalicular diameter	0.78	0.1	0.7	um	above
543-2	current	length per canaliculus	28.9258	25	50	um	inside
543-2	current	canaliculi per lacuna	7.5833	40	115	count	below
543-2	current	areal density	0.343857	0.5	0.9	per um^2	below
543-2	ridge	canalicular diameter	0.937443	0.1	0.7	um	above
543-2	ridge	length per canaliculus	65.715	25	50	um	above
543-2	ridge	canaliculi per lacuna	6.9167	40	115	count	below
543-2	ridge	areal density	0.33848	0.5	0.9	per um^2	below
543_3	current	canalicular diameter	0.78	0.1	0.7	um	above
543_3	current	length per canaliculus	27.0826	25	50	um	inside
543_3	current	canaliculi per lacuna	6.8889	40	115	count	below
543_3	current	areal density	0.347229	0.5	0.9	per um^2	below
543_3	ridge	canalicular diameter	0.937443	0.1	0.7	um	above
543_3	ridge	length per canaliculus	83.0437	25	50	um	above
543_3	ridge	canaliculi per lacuna	7.1111	40	115	count	below
543_3	ridge	areal density	0.345307	0.5	0.9	per um^2	below
543_z13	current	canalicular diameter	0.78	0.1	0.7	um	above
543_z13	current	length per canaliculus	33.1215	25	50	um	inside
543_z13	current	canaliculi per lacuna	6.7273	40	115	count	below
543_z13	current	areal density	0.336942	0.5	0.9	per um^2	below
543_z13	ridge	canalicular diameter	0.937443	0.1	0.7	um	above
543_z13	ridge	length per canaliculus	62.5411	25	50	um	above
543_z13	ridge	canaliculi per lacuna	6.3636	40	115	count	below
543_z13	ridge	areal density	0.330051	0.5	0.9	per um^2	below
682_z08	current	canalicular diameter	0.78	0.1	0.7	um	above
682_z08	current	length per canaliculus	24.1873	25	50	um	below
682_z08	current	canaliculi per lacuna	10	40	115	count	below
682_z08	current	areal density	0.266512	0.5	0.9	per um^2	below
682_z08	ridge	canalicular diameter	0.937443	0.1	0.7	um	above

682_z08	ridge	length per canaliculus	41.7597	25	50	um	inside
682_z08	ridge	canaliculi per lacuna	10.625	40	115	count	below
682_z08	ridge	areal density	0.283281	0.5	0.9	per um^2	below
682_z23	current	canalicular diameter	0.78	0.1	0.7	um	above
682_z23	current	length per canaliculus	20.5961	25	50	um	below
682_z23	current	canaliculi per lacuna	6.7273	40	115	count	below
682_z23	current	areal density	0.294601	0.5	0.9	per um^2	below
682_z23	ridge	canalicular diameter	0.937443	0.1	0.7	um	above
682_z23	ridge	length per canaliculus	45.5301	25	50	um	inside
682_z23	ridge	canaliculi per lacuna	6.8182	40	115	count	below
682_z23	ridge	areal density	0.318926	0.5	0.9	per um^2	below
682_z29	current	canalicular diameter	0.78	0.1	0.7	um	above
682_z29	current	length per canaliculus	19.8912	25	50	um	below
682_z29	current	canaliculi per lacuna	5.5455	40	115	count	below
682_z29	current	areal density	0.288343	0.5	0.9	per um^2	below
682_z29	ridge	canalicular diameter	0.937443	0.1	0.7	um	above
682_z29	ridge	length per canaliculus	62.8354	25	50	um	above
682_z29	ridge	canaliculi per lacuna	5.3636	40	115	count	below
682_z29	ridge	areal density	0.304215	0.5	0.9	per um^2	below

Reporting only: no value below was used to choose any parameter, threshold or scale.

Count per lacuna, canalicular length, volume density and porosity are 3D quantities. A single 2D section is not expected to match them, and nothing here is scaled to make it match.

The width is limited by the optical resolution of the confocal microscope, so it is a width in the image and not a true canalicular diameter.

Several published values are human, not mouse, and come from other preparations and resolutions.

Pixel size 0.13 um/px, rounded, unconfirmed per image, so every micrometre value carries that.

This is a sanity check, not a target.